JFlex group project

ArithLex: an arithmetic input scanner

Reads statements like total=12.5+qty*2; and lists every token with its type, lexeme, line and column

Group 07

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Input format and project scope

Input example

```
total=12.5+qty*2;
```

File: tests\inputs\N01.txt (line 1)

Expected output

```
IDENTIFIER  total  1  1
ASSIGN      =      1  6
DECIMAL     12.5   1  7
...5 more tokens (8 in total)
```

Supported forms
Names, integers, decimals, = + - * / > >= ( ) ;
and whitespace

Outside the scope
Evaluating expressions, checking grammar
and matching parentheses

Token definitions and competing rules

Token type	Pattern or macro	Example lexeme
IDENTIFIER	`{LETTER} ({LETTER} | {DIGIT})*`	total, qty_2
INTEGER	`{DIGIT}+`	12
DECIMAL	`{DIGIT}+ "." {DIGIT}+`	12.5
ASSIGN, PLUS, MINUS, STAR, SLASH	`"="  "+"  "-"  "*"  "/"`	= + - * /
GT, GE	`">"  ">="`	> >=
LPAREN, RPAREN, SEMICOLON	`"("  ")"  ";"`	( ) ;

Two competing-rule examples

12.5: INTEGER matches 2 characters, DECIMAL matches 4, so DECIMAL wins
>=: GT matches 1 character, GE matches 2, so GE wins (longest match)

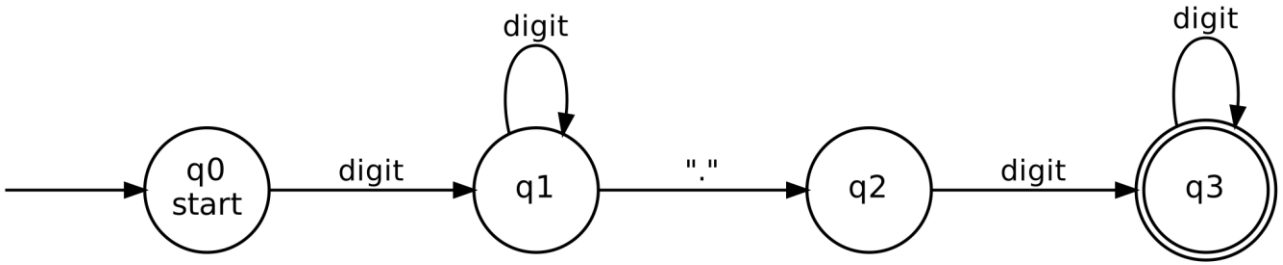
Scanner design and one token automaton

Program structure

Input file → Main.java → ArithScanner → Token rows and errors

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One DFA: the DECIMAL token



Link to the JFlex rule

```
DECIMAL = {DIGIT}+ "." {DIGIT}+
```

Accepted: 12.5 ends in q3
Rejected: 3. stops in q2, so INTEGER 3 and an ERROR

Test evidence and one corrected bug

Test results

Normal: 6 tests, 6 passed
Boundary: 5 tests, 5 passed
Invalid: 4 tests, 4 passed

Total: 15 passed, 0 failed

One bug and its correction

Bug: header and rows did not line up

Cause: columns were separated with tab characters

Fix: String.format with fixed widths, same format for header and rows

Recorded with tests\run_tests.bat (see tests\results_table.md)

Live demonstration

1. Generate and compile

```
C:\ITC\YEAR 4\Semester I\Compilation\Lab01_Group7_ArithScanner>java -jar lib\jflex-full-1.9.1.jar -d src\generated src\arith.jflex
Reading "src\arith.jflex"
Constructing NFA : 49 states in NFA
Converting NFA to DFA :
..................
21 states before minimization, 17 states in minimized DFA
Writing code to "src\generated\ArithScanner.java"

C:\ITC\YEAR 4\Semester I\Compilation\Lab01_Group7_ArithScanner>javac -d build src\generated\ArithScanner.java src\Main.java
```

2. Normal input (N01)

total=12.5+qty*2;

```
C:\ITC\YEAR 4\Semester I\Compilation\Lab01_Group7_ArithScanner>java -cp build Main tests\inputs\N01.txt
TOKEN TYPE   LEXEME       LINE   COLUMN
IDENTIFIER   total        1      1
ASSIGN       =            1      6
DECIMAL      12.5         1      7
PLUS         +            1      11
IDENTIFIER   qty          1      12
STAR         *            1      15
INTEGER      2            1      16
SEMICOLON    ;            1      17
```

3. Competing rule (B02) a>=b>c

```
C:\ITC\YEAR 4\Semester I\Compilation\Lab01_Group7_ArithScanner>java -cp build Main tests\inputs\B02.txt
TOKEN TYPE   LEXEME       LINE   COLUMN
IDENTIFIER   a            1      1
GE           >=           1      2
IDENTIFIER   b            1      4
GT           >            1      5
IDENTIFIER   c            1      6
```

4. Competing rule (B01) 12.5

```
C:\ITC\YEAR 4\Semester I\Compilation\Lab01_Group7_ArithScanner>java -cp build Main tests\inputs\B01.txt
TOKEN TYPE   LEXEME       LINE   COLUMN
DECIMAL      12.5         1      1
```

5. Invalid input (E01) 7 @ 8

```
C:\ITC\YEAR 4\Semester I\Compilation\Lab01_Group7_ArithScanner>java -cp build Main tests\inputs\E01.txt
TOKEN TYPE   LEXEME       LINE   COLUMN
INTEGER      7            1      1
ERROR        @            1      3
INTEGER      8            1      5
```

6. All tests passing

```
C:\ITC\YEAR 4\Semester I\Compilation\Lab01_Group7_ArithScanner>tests\run_tests.bat
PASS  B01.txt
PASS  B02.txt
PASS  B03.txt
PASS  B04.txt
PASS  B05.txt
PASS  E01.txt
PASS  E02.txt
PASS  E03.txt
PASS  E04.txt
PASS  N01.txt
PASS  N02.txt
PASS  N03.txt
PASS  N04.txt
PASS  N05.txt
PASS  N06.txt

Passed: 15   Failed: 0
```

Results and team contributions

What works

All 15 tests pass
Limitation: no grammar check, so 3 + ; is accepted
Lesson: longest match and rule order decide every conflict

Who contributed what

TyKun: arith.jflex rules and tokens
Daraleap: Main.java, build, README
Sopanha: tests and results table
Makara: report, DFA and slides

Longest match and rule order drive every scanner decision